Introduction to EXP2-1 Part 2

EXP2-1 Same or Different?

Semester B, Weeks 1-3

The **aim** of EXP2-1 is to develop tools and skills, in order to:

- Present and report data in the correct way

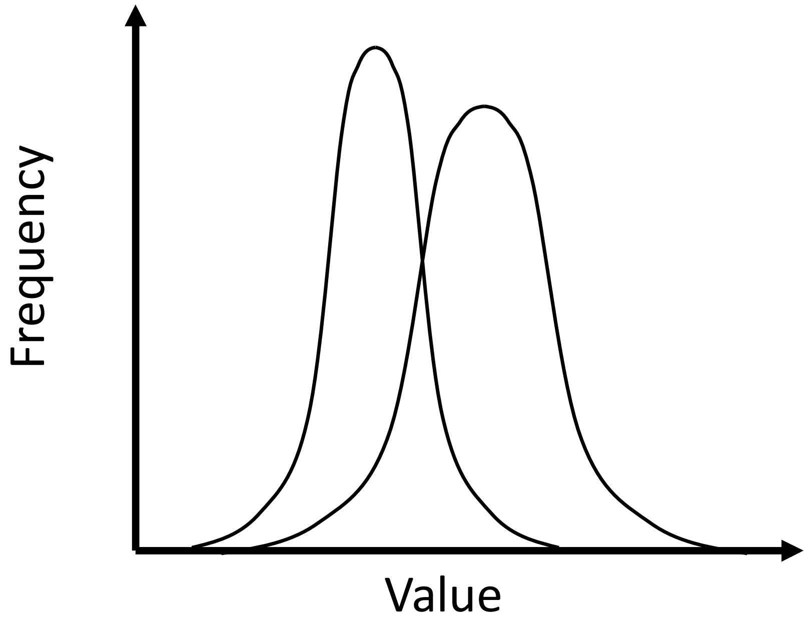
  (units, significant figures, error, and others)
- Determine whether the data are statistically significant

Part 2 Significance tests

(comparing results, or samples)

Introduction to EXP2-1 Part 2

Significance tests



Are these 2 sets of data different from each other?

Significance tests

3 ways to determine differences between sets of data that vary randomly:

z-test - sample different from a population

t-test - samples different from each other

F-test - samples have different variance from each other

Some definitions

Null hypothesis (H_0) : There is NO difference / NO effect / NO relationship between the target groups.

Alternative hypothesis (H_1): There IS difference / effect / relationship between the target groups.

The opposite region from H_0

Significance tests are used to test the hypothesis, and the results of the test would reveal whether the hypothesis would be rejected or supported, providing you the answer to your research question.

Two-tailed: To test whether the difference exists.

One-tailed test: To test the direction of the difference.

The z-test

Is a sample different from a known population?

You can knowledge about a population from a previous sample, if you had enough data.

e.g. Are your measured values different from known values?

Used to compare new values to **a <u>large</u> existing data set**

Used in process control to see if a process has changed

Formula – z standard score

z standard score: a measure of how many standard deviations

a sample mean is from the known population mean.

$$z = \frac{\bar{X} - \mu}{\sigma_{\bar{X}}} = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

z > 0, sample mean > population mean;
z ~ 0, sample mean ~ population mean;
z < 0, sample mean < population mean.

$\bar{X}$: *mean of sample* $\sigma_{\bar{X}}$: SE *of sampling distribution*
μ : *mean of population* S: *SD of sample (n>30)*

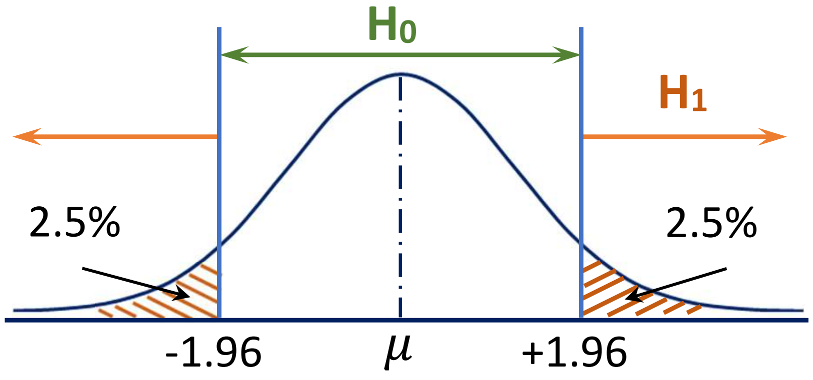
Significance level (α): the probability of rejecting the null hypothesis.

α = 0.05, 5% level; z = ±1.96 (2-tailed), z= ±1.64 (1-tailed)

α = 0.01, 1% level; z = ±2.58 (2-tailed), z= ±2.33 (1-tailed)

1 & 2 tailed z-tests

Two-tailed test: $\alpha = 0.05, z_{critical} = z_{\alpha/2} = \pm 1.96$

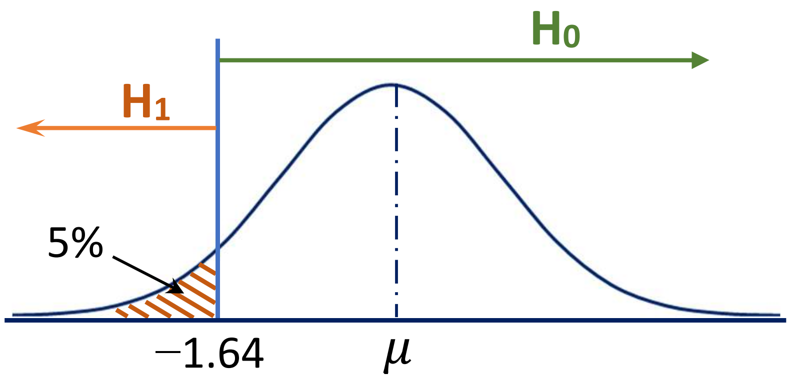


H_0: $\bar{X} = \mu$, H_1: $\bar{X} \neq \mu$

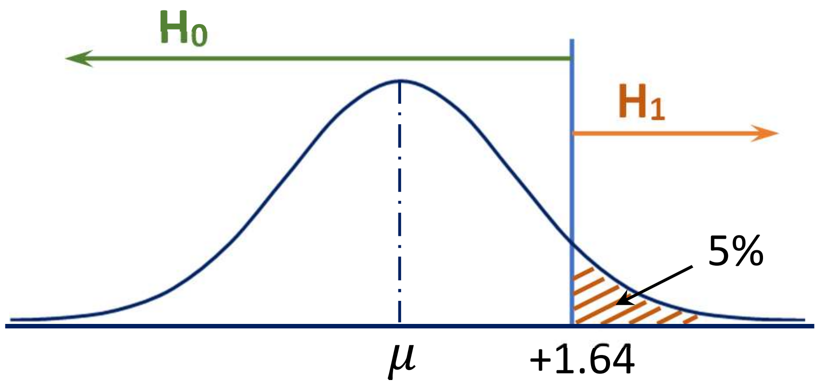
$|z| < |z_{\alpha/2}|$, *accept* H_0

$|z| > |z_{\alpha/2}|$, *reject* H_0

One-tailed test: $\alpha = 0.05, z_{critical} = z_{\alpha} = \pm 1.64$



$H_0 : \bar{X} > \mu$, H_1: $\bar{X} \leq \mu$



H_0: $\bar{X} < \mu$, H_1: $\bar{X} \geq \mu$

The z-test

<u>Null hypothesis</u> (H_0):

The means of the sample and population are the same.

Set a significance level, α = 0.05, z = 1.96 (critical z)

i.e. 95 times out of 100 your hypothesis is correct

If more than 5% chance of being wrong, reject the null hypothesis

If z < 1.96, **H_0** is correct, sample and population are the same

If z > 1.96, **H_0** is incorrect, it is “statistically significant”

(“we can gain confidence”) that sample and population are different

WE CAN ONLY TEST the Null Hypothesis

Exercise 1

The average standard service life for an electronic device is 1200 hours with standard deviation σ = 135. A factory announced that they have applied a new advanced processing method which could improve the current standard. 25 samples have been selected with an average service life of 1250 hours.

Is there a significant difference between the service life for newly produced devices and the original standard ones?

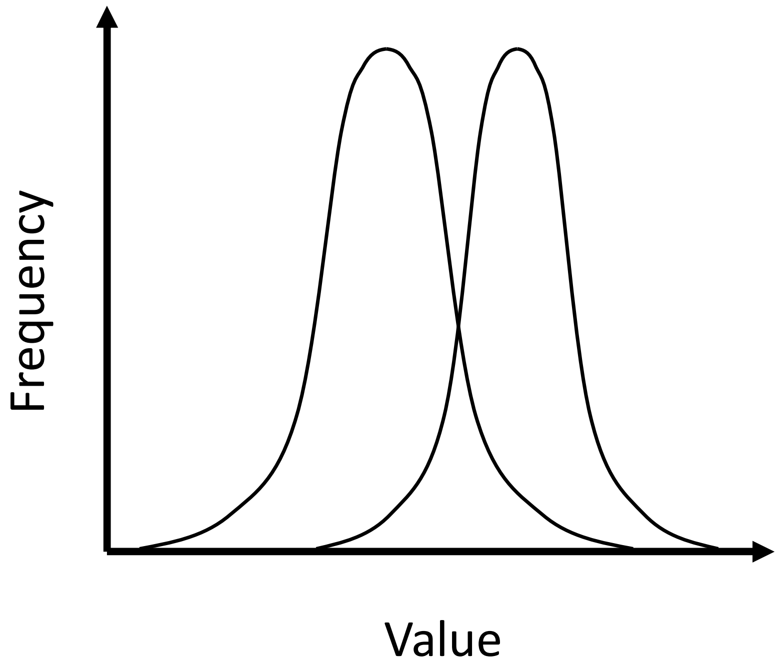
Is the service life of newly produced devices significantly higher than the standard ones?

Z-test conditions

- Compare whether there is significant difference between the mean values of two samples (normally for large samples, n>30);
- Compare whether there is significant difference between a new data and a population when the SD of the population is known.

Student’s t-test

Are two samples different from each other?



e.g. Are data from a control group and experimental group different?

Has a pharmaceutical drug had an effect?

Are men and women equally tall?

Tests

Comparing different types of variables

- One **numerical** variable (e.g. height)
- One **categorical** variable (groups A or B, women or men, etc.)
- Comparisons of means, medians and distributions
- **t tests**, Mann-Whitney, ANOVA

Two **categorical** variables:

- Cross-tabulation & proportions
- Chi-squared & McNemar tests

Two **numerical** variables:

- Scatter plots
- Regression & correlation

Student’s t-test

Research questions:

- Does the **typical** value of the **numerical variable** depend on the value of the **categorical variable**?
- Do the samples **differ** systematically in **mean** or **median** value?

There are MANY t-tests types, each one with their own equation: make sure you apply the one corresponding to the case you are studying

Two-sample t-test

Independent samples subjected to same measurement.

- **Purpose:** Test significance of **difference in means** of 2 samples of **numerical** variables. Samples refer to 2 different values of a **categorical** variable.

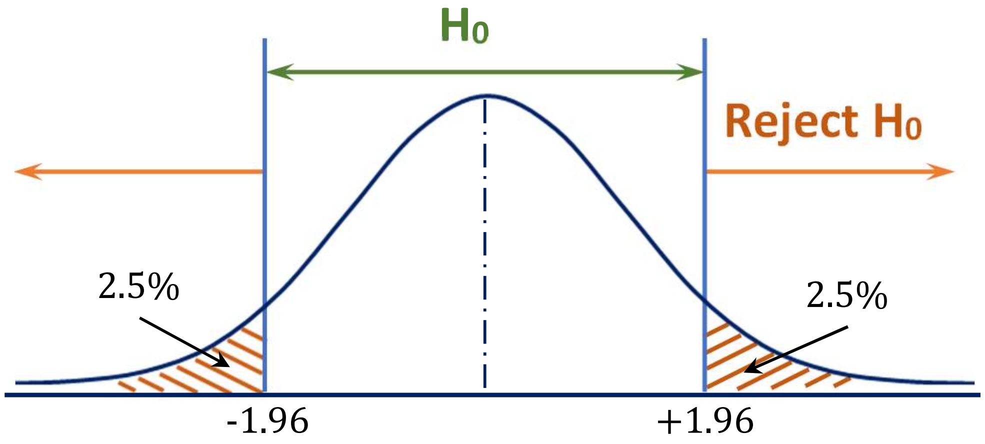
  e.g. Are men and women equally tall?

- **Null hypothesis (H_0):** There is no difference between the sample 1&2.

  There is **no** systematic **difference**. Sample means differ only by **chance**.

- **Alternative hypothesis** (**H_1**): The sample means are different.
  - Men and women are not equally tall (but we're not saying who is taller) (**2-tailed**)
  - Men are taller than women (**1-tailed**) OR
  - Women are taller than men (**1-tailed**)

1 and 2 tailed tests

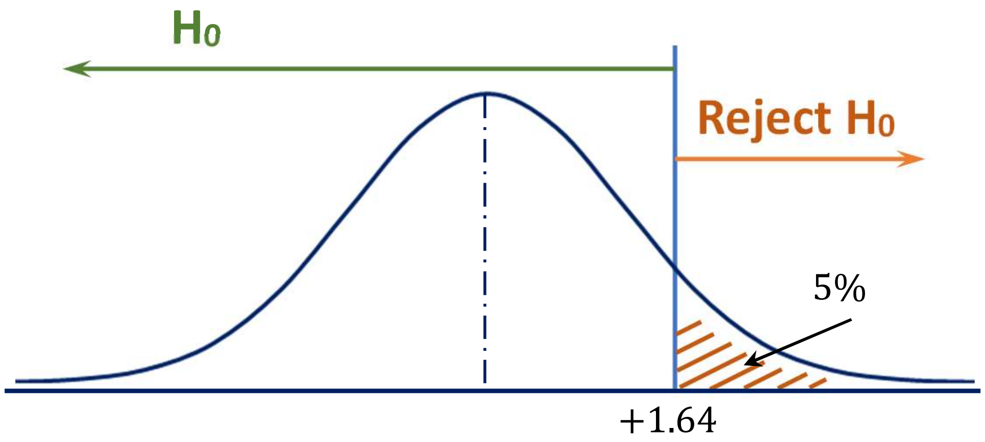


Two-tailed test at the 5% level

$\alpha = 0.05, z = 1.96$ (95% $confidence$)

2-tailed: difference can be in either direction (or distribution tail)

$$H_0\text{: } \mu_1 = \mu_2$$



One-tailed test at the 5% level

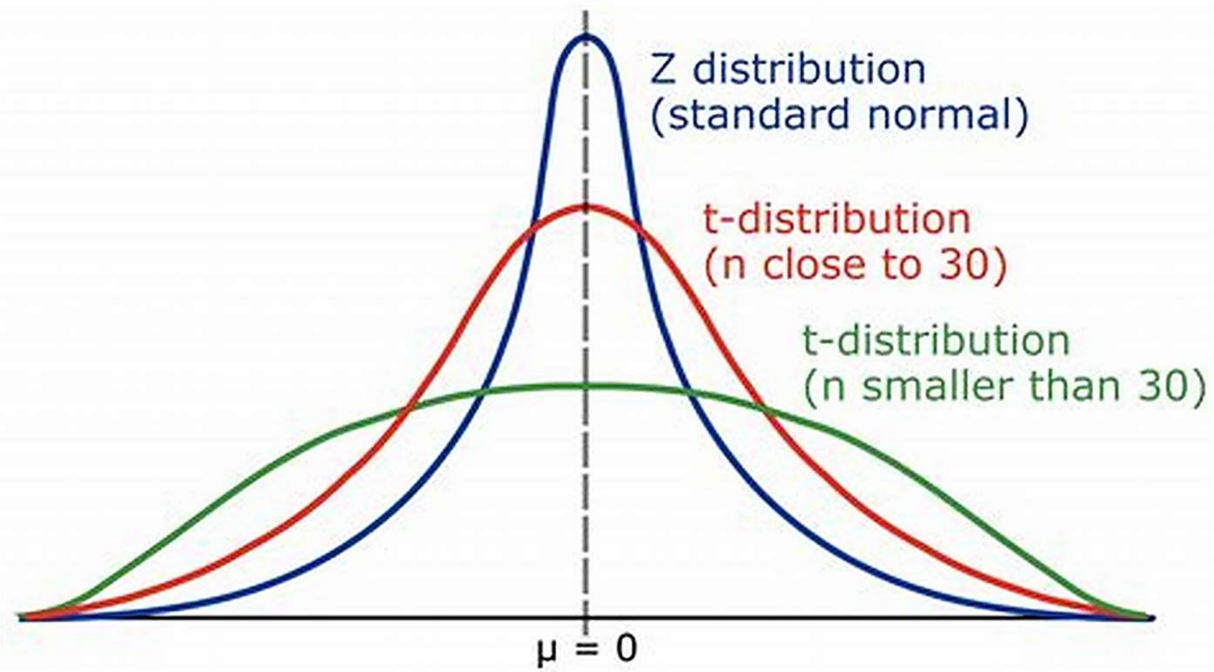
1-tailed: difference can only be in one direction (or distribution tail)

$$H_0\text{: } \mu_1 > \mu_2 \text{ or } \mu_1 < \mu_2$$

Two-sample t-test

How 2-sample (independent) t-test works?

- Calculate test statistic $d/SE_{(d)}$ (“**sample t value**”) and compare with a critical t-value for the degree of freedom of the sample (related to number of measurements or data points)
- If $\mathbf{H_0}$ is true, this test statistic follows the t distribution (like a normal distribution but tails are slightly stretched out)



Formula – t statistic

t statistic value: a standardized value that is calculated from sample data for a hypothesis test.

$$t = \frac{\overline{X}_1 - \overline{X}_2}{\sqrt{(s_1^2/n_1) + (s_2^2/n_2)}}$$

<u>Independent samples & Equal variances</u>

$\overline{X_1}$: $mean\ of\ sample\ 1$

$\overline{X_2}$: $mean\ of\ sample\ 2$

S_1: $SD\ of\ sample\ 1$

S_2: $SD\ of\ sample\ 2$

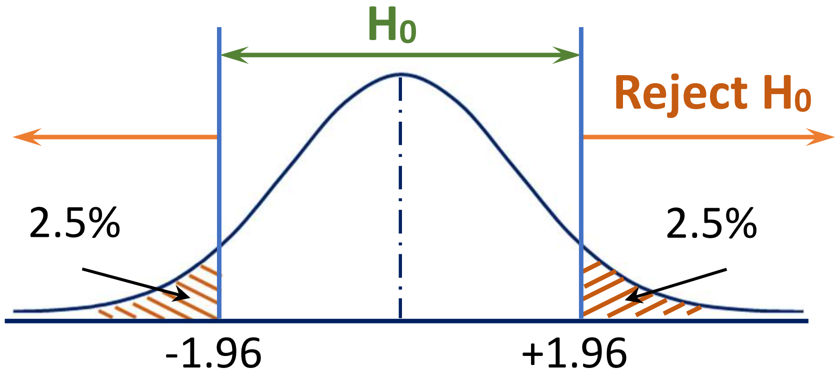
n_1: $Size\ of\ sample\ 1$

n_2: $Size\ of\ sample\ 2$

Degree of freedom v: shows how many 'free' data points are available in your test for making comparisons.

1 & 2 tailed t-tests

Two-tailed test: $\alpha = 0.05, t_{critical} = t_{\left(1-\frac{1}{2}\alpha,\nu\right)}$

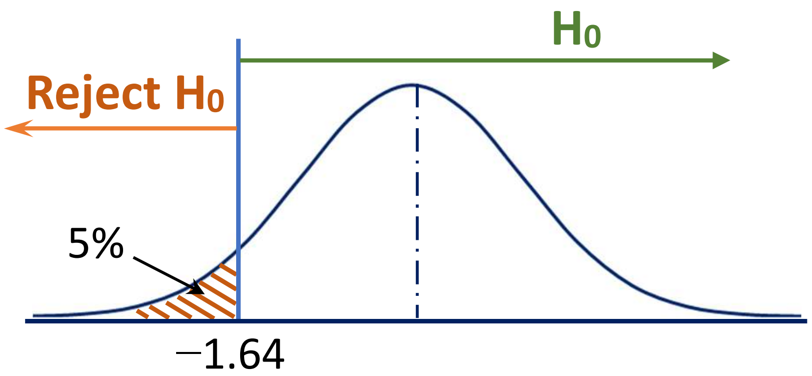


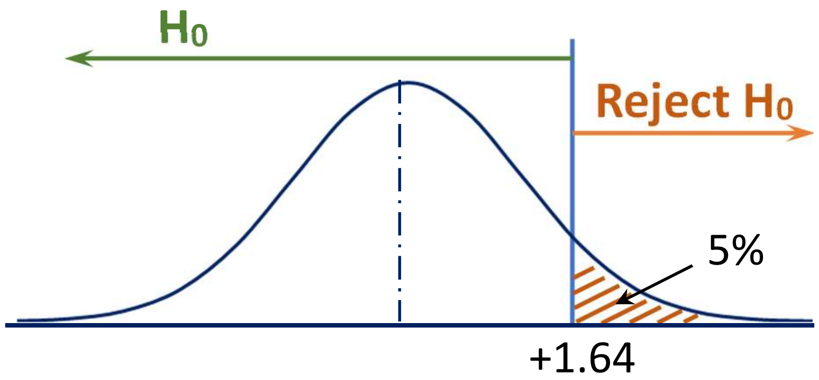
$H_0: \mu_1 = \mu_2$

$t < t_{critical}, accept\ H_0$

$t > t_{critical}, reject\ H_0$

One-tailed test: $\alpha = 0.05, t_{critical} = t_{(1-\alpha,\nu)}$





$H_0: \mu_1 > \mu_2\ or\ \mu_1 < \mu_2$ $\quad t < t_{critical}\ or\ , t > t_{critical}, reject\ H_0$

Critical Values of the Student's t Distribution

Taken from: https://www.itl.nist.gov/div898/handbook/eda/section3/eda3672.htm

The table in following pages contains critical values of the Student’s t distribution. The t distribution is symmetric so that $\mathbf{t_{1-\alpha,\nu} = -t_{\alpha,\nu}}$

The t table can be used for both one-tailed (also called one-sided), either lower or upper, and two-tailed (also two-sided) tests using the appropriate value of α.

The difference lies in finding out whether you are conducting a one-tailed test, or two-tailed test. *For a one-tailed test, use the 1-α column* (for example “0.95” for 95% confidence, α=0.05); *for a two-tailed test, use the 1-α/2 column* (for example “0.975” for 95% confidence, α=0.05) vs **the degrees of freedom for equal variance (= number of events-2)** for the t-critical values.

Decide whether you will be conducting a one-tailed or two-tailed test. Calculate the t statistic for your samples. Find the critical t value and compare. If t > critical t, then you reject the null hypothesis H_0, gaining confidence in your alternative hypothesis.

Critical values of Student's *t* distribution with *v* degrees of freedom

Probability less than the critical value ($t_{1-\alpha,v}$)

v	0.90	0.95	0.975	0.99	0.995	0.999
1.	3.078	6.314	12.706	31.821	63.657	318.313
2.	1.886	2.920	4.303	6.965	9.925	22.327
3.	1.638	2.353	3.182	4.541	5.841	10.215
4.	1.533	2.132	2.776	3.747	4.604	7.173
5.	1.476	2.015	2.571	3.365	4.032	5.893
6.	1.440	1.943	2.447	3.143	3.707	5.208
7.	1.415	1.895	2.365	2.998	3.499	4.782
8.	1.397	1.860	2.306	2.896	3.355	4.499
9.	1.383	1.833	2.262	2.821	3.250	4.296
10.	1.372	1.812	2.228	2.764	3.169	4.143
11.	1.363	1.796	2.201	2.718	3.106	4.024
12.	1.356	1.782	2.179	2.681	3.055	3.929
13.	1.350	1.771	2.160	2.650	3.012	3.852
14.	1.345	1.761	2.145	2.624	2.977	3.787
15.	1.341	1.753	2.131	2.602	2.947	3.733
16.	1.337	1.746	2.120	2.583	2.921	3.686
17.	1.333	1.740	2.110	2.567	2.898	3.646
18.	1.330	1.734	2.101	2.552	2.878	3.610
19.	1.328	1.729	2.093	2.539	2.861	3.579
20.	1.325	1.725	2.086	2.528	2.845	3.552
21.	1.323	1.721	2.080	2.518	2.831	3.527
22.	1.321	1.717	2.074	2.508	2.819	3.505
23.	1.319	1.714	2.069	2.500	2.807	3.485

Sample I: n_1=7

Sample II: n_2=4

$$\alpha = 0.05$$

$$1 - \alpha = 0.95$$

$$\textit{Degrees of freedom}$$

$$= n_1 + n_2 - 2 = 9$$

$$1 - \textit{tailed test}$$

$$t_{critical} = 1.833$$

Exercise 2

There are two processing methods to produce a type of electronic device.

$$\text{Method 1: } \overline{X_1} = 1200 \text{ hours}, S_1 = 135 \text{ hours}, n_1 = 36$$

$$\text{Method 2: } \overline{X_2} = 1250 \text{ hours}, S_2 = 120 \text{ hours}, n_2 = 42$$

Is there a significant difference between the service life of the devices produced from the two methods? (α=0.05)

You can find the t critical value here:
https://www.itl.nist.gov/div898/handbook/eda/section3/eda3672.htm

T-test conditions

t-test **assumes** that the sample means follows a **normal distribution** when $\mathbf{H_0}$ is true:

- Valid if **population is normal distribution** (in which case samples will appear approximately normal)
- Also valid for **non-normal** populations (and samples) **if** sample size is **large** (> 30 per sample)
- Should **not** be used if samples are both <u>small & non-normal</u>.
- Unreliable if samples are too small to tell whether normally distributed.

Summary

z-test **-** sample different from a population

(are the new results different from the 'norm'?)

t-test **-** samples different from each other

(are two samples different from the each other?)

EXP2-1 Schedule – P Groups

Date:	Time:	Place:	Activity:
Monday 17th	08:30-10:10	QMES A310	Introduction to QXU5017 Introduction to EXP2-1 Part1
Wednesday 19th	08:30-10:10		Group Meeting 1 (Review and check the raw data)
Friday 21st	10:30-12:10		Group Meeting 2 (Write Report Section 1-4)
Monday 24th	08:30-10:10	QMES A310	Introduction to EXP2-1 Part2
Wednesday 26th	08:30-10:10		Group Meeting 3 (Different sample comparison)
Friday 28th	10:30-12:10		Group Meeting 4 (Complete report Section 5-8)
Wednesday 5th	08:30-10:10	A310	Report submission

EXP2-1 Schedule – M Groups

Date:	Time:	Place:	Activity:
Monday 17th	10:30-12:10	QMES A310	Introduction to QXU5017 Introduction to EXP2-1 Part1
Wednesday 19th	10:30-12:10		Group Meeting 1 (Review and check the raw data)
Friday 21st	08:30-10:10		Group Meeting 2 (Write Report Section 1-4)
Monday 24th	10:30-12:10	QMES A310	Introduction to EXP2-1 Part2
Wednesday 26th	10:30-12:10		Group Meeting 3 (Different sample comparison)
Friday 28th	08:30-10:10		Group Meeting 4 (Complete report Section 5-8)
Wednesday 5th	10:30-12:10	A310	Report submission